UMRRA1 T166 B AUTOMOTIVE V1.0.0 USER INTERFACE

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UMRRA1 T166 B AUTOMOTIVE

USER INTERFACE VERSION:
v1.0.0

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CONTENT

1 COMMUNICATION DATA STREAM SERVICE 2
1.1 COMTARGETLIST PORT 2
2 LEGAL DISCLAIMER NOTICE 5
APPENDIX 6
A COMMUNICATION DATA SERVICE API 6
B USER INTERFACE INSTRUCTIONS UMRRA1 T166 B AUTOMOTIVE VERSION 1.0.0 10
B.1 Parameter Section auto_interface_rrm 10
B.2 Status Section auto_interface_rrm 11
B.3 Command Section auto_interface_rrm_command 13

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1 COMMUNICATION DATA STREAM SERVICE

With the communication data stream service smartmicro ports can be received as C++ objects with simplified access functions, which are generated by the user interface. Smartmicro ports are data buffers which contains data recorded by the radar data: e.g objects, statistics, statuses of device etc. Each port contains a generic port header, with a port description : version, id, size etc. Sometime ports also contains dynamic list of objects. In order to receive a port, a callback needs to be registered with the service. The callback will be carried out periodically every sensor cycle time.

Please note:

- This callback will be called in the context of a receiver thread, so the data needs to be copied and the callback must be released. Otherwise, the reception will be blocked.
- It is possible to use one callback function for several clients with the same port and same user interface, but it is not allowed to use one callback function for different ports or different user interfaces.

For more details please see the examples below. The following ports are supported.

1.1 COMTARGETLIST PORT

Description:
To receive the port called "ComTargetList" from a specific client, please use the following registration interface:

```
#include <umrra1_t166_b_automotive_v1_0_0/DataStreamServiceIface.h>

void ReceiveComTargetListClbk(IN std::shared_ptr<com::master::umrra1_t166_b_automotive_v1_0_0::
    comtargetlist::ComTargetList> comTargetList, com::types::ClientId clientId)
{
        // Getting members of  ComTargetList
    std::shared_ptr<com::master::umrra1_t166_b_automotive_v1_0_0::comtargetlist::PortHeader>
        portHeader =
                        comTargetList->GetPortHeader();
    std::shared_ptr<com::master::umrra1_t166_b_automotive_v1_0_0::comtargetlist::TargetListHeader
        > targetListHeader =
                        comTargetList->GetTargetListHeader();
    auto targetList = comTargetList->GetTargetList();
    // Getting members of  PortHeader
    std::cout << "Variable PortIdentifier:"
              << portHeader->GetPortIdentifier()
              << std::endl;
    std::cout << "Variable PortVersionMajor:"
              << portHeader->GetPortVersionMajor()
              << std::endl;
    std::cout << "Variable PortVersionMinor:"
              << portHeader->GetPortVersionMinor()
              << std::endl;
    std::cout << "Variable Timestamp:"
              << portHeader->GetTimestamp()
              << std::endl;
    std::cout << "Variable PortSize:"
              << portHeader->GetPortSize()
              << std::endl;
    std::cout << "Variable BodyEndianness:"
              << portHeader->GetBodyEndianness()
              << std::endl;
    std::cout << "Variable PortIndex:"
              << portHeader->GetPortIndex()
              << std::endl;
    std::cout << "Variable HeaderVersionMajor:"
              << portHeader->GetHeaderVersionMajor()
              << std::endl;
    std::cout << "Variable HeaderVersionMinor:"
              << portHeader->GetHeaderVersionMinor()
```


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```
            << std::endl;
    // Getting members of  TargetListHeader
    std::cout << "Variable CycleTime:"
            << targetListHeader->GetCycleTime()
            << std::endl;
    std::cout << "Variable NumberOfTargets:"
            << targetListHeader->GetNumberOfTargets()
            << std::endl;
    std::cout << "Variable AcquisitionTxAntIdx:"
            << targetListHeader->GetAcquisitionTxAntIdx()
            << std::endl;
    std::cout << "Variable AcquisitionSweepIdx:"
            << targetListHeader->GetAcquisitionSweepIdx()
            << std::endl;
    std::cout << "Variable AcquisitionCfIdx:"
            << targetListHeader->GetAcquisitionCfIdx()
            << std::endl;
    std::cout << "Variable Prf:"
            << targetListHeader->GetPrf()
            << std::endl;
    std::cout << "Variable UmambiguousSpeed:"
            << targetListHeader->GetUmambiguousSpeed()
            << std::endl;
    std::cout << "Variable AcquisitionStart:"
            << targetListHeader->GetAcquisitionStart()
            << std::endl;
    // Getting members of  Target
    for(auto& target : targetList)
    {
        std::cout << "Variable Range:"
                << target->GetRange()
                << std::endl;
        std::cout << "Variable SpeedRadial:"
                << target->GetSpeedRadial()
                << std::endl;
        std::cout << "Variable AzimuthAngle:"
                << target->GetAzimuthAngle()
                << std::endl;
        std::cout << "Variable ElevationAngle:"
                << target->GetElevationAngle()
                << std::endl;
        std::cout << "Variable VarianceRange:"
                << target->GetVarianceRange()
                << std::endl;
        std::cout << "Variable VarianceSpeed:"
                << target->GetVarianceSpeed()
                << std::endl;
        std::cout << "Variable VarianceAzimuthAngle:"
                << target->GetVarianceAzimuthAngle()
                << std::endl;
        std::cout << "Variable VarianceElevationAngle:"
                << target->GetVarianceElevationAngle()
                << std::endl;
        std::cout << "Variable Rcs:"
                << target->GetRcs()
                << std::endl;
        std::cout << "Variable FalseAlarmProbability:"
                << target->GetFalseAlarmProbability()
                << std::endl;
        std::cout << "Variable Flags:"
                << target->GetFlags()
                << std::endl;
        std::cout << "Variable Power:"
                << target->GetPower()
                << std::endl;
        std::cout << "Variable Noise:"
                << target->GetNoise()
```


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```
                << std::endl;
        std::cout << "Variable PeakIdx:"
                << target->GetPeakIdx()
                << std::endl;
    }

}
    auto comDataStreamServ = com::master::umrra1_t166_b_automotive_v1_0_0::DataStreamServiceIface
        ::Get();
    ClientId clientIdA = 1024; // client id from sensor a
    ClientId clientIdB = 1025; // client id from sensor b
    ReceiveComTargetListCallback callback =
                        std::bind(&ReceiveComTargetListClbk,
                        std::placeholders::_1,
                        std::placeholders::_2);
    if(ERROR_CODE_OK != comDataStreamServ->RegisterComTargetListReceiveCallback(clientIdA,
        callback)
    {
        std::cout << "Failed to register ComTargetList port callback" << std::endl;
    }
    if(ERROR_CODE_OK != comDataStreamServ->RegisterComTargetListReceiveCallback(clientIdB,
        callback))
    {
        std::cout << "Failed to register ComTargetList port callback" << std::endl;
    }
```

For a more detailed API description, please see Appendix A.

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A COMMUNICATION DATA SERVICE API

A **ComTargetList**
Description: Target list port for automotive standard products.
The object ComTargetList provides the following APIs:

```
std::shared_ptr<com::master::umrra1_t166_b_automotive_v1_0_0::PortHeader> GetPortHeader()
    const;
```

Returns pointer to PortHeader object, whose access functions are described below:
A **PortHeader**
Description: smartmicro Generic Port Header
The object PortHeader provides the following APIs:

```
uint32_t GetPortIdentifier() const;
```

Returns value of PortIdentifier of uint32_t data type.
PortIdentifier - Unique port identifier

```
int16_t GetPortVersionMajor() const;
```

Returns value of PortVersionMajor of int16_t data type.
PortVersionMajor - Major port version

```
int16_t GetPortVersionMinor() const;
```

Returns value of PortVersionMinor of int16_t data type.
PortVersionMinor - Minor port version

```
uint64_t GetTimestamp() const;
```

Returns value of Timestamp of uint64_t data type.
Timestamp - A timestamp

```
uint32_t GetPortSize() const;
```

Returns value of PortSize of uint32_t data type.
PortSize - Port size (includes this header)

```
uint8_t GetBodyEndianness() const;
```

Returns value of BodyEndianness of uint8_t data type.
BodyEndianness - Port body endianness[1:big endian][2:little endian]

```
uint8_t GetPortIndex() const;
```

Returns value of PortIndex of uint8_t data type.
PortIndex - To identify multiple instances of the same port

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```
uint8_t GetHeaderVersionMajor() const;
```

Returns value of HeaderVersionMajor of uint8_t data type.
HeaderVersionMajor - Generic port header major version

```
uint8_t GetHeaderVersionMinor() const;
```

Returns value of HeaderVersionMinor of uint8_t data type.
HeaderVersionMinor - Generic port header minor version

```
std::shared_ptr<com::master::umrra1_t166_b_automotive_v1_0_0::TargetListHeader>
    GetTargetListHeader() const;
```

Returns pointer to TargetListHeader object, whose access functions are described below:
A **TargetListHeader**
Description: Static header of the target list
The object TargetListHeader provides the following APIs:

```
float32_t GetCycleTime() const;
```

Returns value of CycleTime of float32_t data type.
CycleTime - Cycle Time

```
uint16_t GetNumberOfTargets() const;
```

Returns value of NumberOfTargets of uint16_t data type.
NumberOfTargets - Number of valid targets in the current target list

```
uint8_t GetAcquisitionTxAntIdx() const;
```

Returns value of AcquisitionTxAntIdx of uint8_t data type.
AcquisitionTxAntIdx - Active TX-antenna index during data acquisition

```
uint8_t GetAcquisitionSweepIdx() const;
```

Returns value of AcquisitionSweepIdx of uint8_t data type.
AcquisitionSweepIdx - Active sweep index during data acquisition

```
uint8_t GetAcquisitionCfIdx() const;
```

Returns value of AcquisitionCfIdx of uint8_t data type.
AcquisitionCfIdx - Active center frequency index during data acquisition

```
uint8_t GetPrf() const;
```

Returns value of Prf of uint8_t data type.
Prf - Used PRF Setting during acquisition, index/set: 0..2 first set; 3..5 second set

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```
float32_t GetUmambiguousSpeed() const;
```

Returns value of UmambiguousSpeed of float32_t data type.
UmambiguousSpeed - Typical value of speed misinterpreations in this cycle

```
uint64_t GetAcquisitionStart() const;
```

Returns value of AcquisitionStart of uint64_t data type.
AcquisitionStart - Start of radar acquisition for target list (NTP coded)

```
const std::vector<std::shared_ptr<com::master::umrra1_t166_b_automotive_v1_0_0::Target>>&
    GetTargetList() const;
```

Returns pointer to array of Target objects, whose access functions are described below:
A **Target**
Description: Represent a single target
The object Target provides the following APIs:

```
float32_t GetRange() const;
```

Returns value of Range of float32_t data type.
Range - Range, distance between sensor and target

```
float32_t GetSpeedRadial() const;
```

Returns value of SpeedRadial of float32_t data type.
SpeedRadial - Radial speed of the target

```
float32_t GetAzimuthAngle() const;
```

Returns value of AzimuthAngle of float32_t data type.
AzimuthAngle - Azimuth angle of the target

```
float32_t GetElevationAngle() const;
```

Returns value of ElevationAngle of float32_t data type.
ElevationAngle - Elevation angle of the target

```
float32_t GetVarianceRange() const;
```

Returns value of VarianceRange of float32_t data type.
VarianceRange - Variance of the range.

```
float32_t GetVarianceSpeed() const;
```

Returns value of VarianceSpeed of float32_t data type.
VarianceSpeed - Variance of the speed.

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```
float32_t GetVarianceAzimuthAngle() const;
```

Returns value of VarianceAzimuthAngle of float32_t data type.
VarianceAzimuthAngle - Variance of the azimuth angle.

```
float32_t GetVarianceElevationAngle() const;
```

Returns value of VarianceElevationAngle of float32_t data type.
VarianceElevationAngle - Variance of the elevation angle.

```
float32_t GetRcs() const;
```

Returns value of Rcs of float32_t data type.
Rcs - Radar cross-section

```
float32_t GetFalseAlarmProbability() const;
```

Returns value of FalseAlarmProbability of float32_t data type.
FalseAlarmProbability - Probability of the false alarm.

```
uint32_t GetFlags() const;
```

Returns value of Flags of uint32_t data type.
Flags - Flags as bitwise OR operation of port_target_list_target_flags 'target flags'.

```
float32_t GetPower() const;
```

Returns value of Power of float32_t data type.
Power - Power

```
float32_t GetNoise() const;
```

Returns value of Noise of float32_t data type.
Noise - Noise

```
uint16_t GetPeakIdx() const;
```

Returns value of PeakIdx of uint16_t data type.
PeakIdx - Index of the peak in the peak list port (PLP) port that this target has been generated from.

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B USER INTERFACE INSTRUCTIONS UMRRA1 T166 B AUTOMOTIVE VERSION 1.0.0

B.1 Parameter Section auto_interface_rrm

Automotive user interface parameters

Parameter Name	center_frequency_idx
Description	Index of center frequency (76125, 76371, 76605, 76845 MHz)
Data Type	u8
Dimensions	None
Access	RW
Default	0
Min	0
Max	3

Parameter Name	frequency_sweep_idx
Description	Index of sweep (0=LRR, 1=MRR, 2=SRR)
Data Type	u8
Dimensions	None
Access	RW
Default	0
Min	0
Max	2

Parameter Name	detection_method
Description	Select detection method (0=Off, 1=Hires)
Data Type	u8
Dimensions	None
Access	RW
Default	1
Min	0
Max	1

Parameter Name	output_control_target_list_eth
Description	Send raw targets via Ethernet, 0 = disabled, 1 = enabled
Data Type	u8
Dimensions	None
Access	RW
Default	1
Min	0
Max	1

Parameter Name	ip_source_address
Description	IP source address (32bit)
Data Type	u32
Dimensions	None
Access	RW
Default	3232238347

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Parameter Name	subnet_mask
Description	Subnet mask (32bit)
Data Type	u32
Dimensions	None
Access	RW
Default	4294967040

Parameter Name	ip_dest_address
Description	IP destination address (32bit)
Data Type	u32
Dimensions	None
Access	RW
Default	3232238337

Parameter Name	ip_dest_port
Description	IP destination port
Data Type	u16
Dimensions	None
Access	RW
Default	55555

B.2 Status Section auto_interface_rrm

customer status section

Status Name	auto_interface_version_major
Description	Automotive interface version major. Increased, if new version is not totally backward compatible.
Data Type	u32
Dimensions	None
Access	R

Status Name	auto_interface_version_minor
Description	Automotive interface version minor. Increased, if parameters or statuses are changed or added. The new version is still backward compatible.
Data Type	u32
Dimensions	None
Access	R

Status Name	sw_generation
Description	Software Version generation
Data Type	u16
Dimensions	None
Access	R

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Status Name	sw_version_major
Description	Software Version major
Data Type	u16
Dimensions	None
Access	R

Status Name	sw_version_minor
Description	Software Version minor
Data Type	u16
Dimensions	None
Access	R

Status Name	sw_version_patch
Description	Software Version patch
Data Type	u16
Dimensions	None
Access	R

Status Name	customer_id
Description	Customer Identifier
Data Type	u16
Dimensions	None
Access	R

Status Name	product_serial
Description	32Bit product id serial
Data Type	u32
Dimensions	None
Access	R

Status Name	product_gen
Description	product generation
Data Type	u32
Dimensions	None
Access	R

Status Name	product_mod_high
Description	product modification high
Data Type	u32
Dimensions	None
Access	R

Status Name	product_mod_low
Description	product modification low
Data Type	u32
Dimensions	None
Access	R

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Status Name	product_rev
Description	product revision
Data Type	u32
Dimensions	None
Access	R

Status Name	u32_systime_low_dword
Description	time in us since power on
Data Type	u32
Dimensions	None
Access	R

Status Name	u32_systime_high_dword
Description	time in us since power on
Data Type	u32
Dimensions	None
Access	R

Status Name	u32_crc32_val
Description	crc value
Data Type	u32
Dimensions	None
Access	R

Status Name	u32_crc_calc_done
Description	crc calculation done
Data Type	u32
Dimensions	None
Access	R

B.3 Command Section auto_interface_rrm_command

Maintain compatible section 1000 commands

Command Name	sensor_reboot
Description	Performs sensor reboot.

Command Name	soft_sensor_reset
Description	Performs soft reset.

Command Name	restore_default_param
Description	Restore default values in RAM and EEPROM.

Command Name	sw_crc_check
Description	Perform crc check

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